机械制造基础 II

周计明

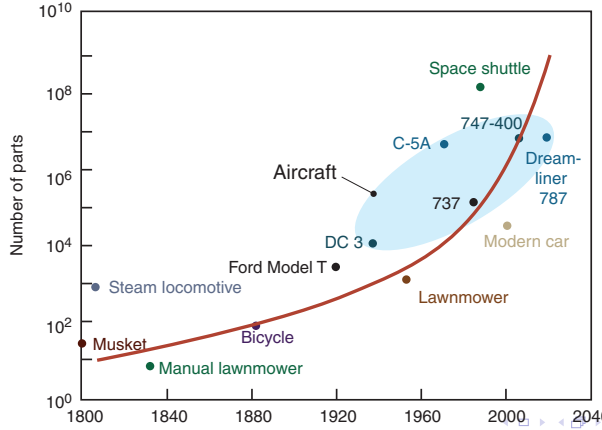
西北工业大学

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Chapter 2: 绪论

制造 Manufacturing

- The vast majority of objects around us consist of numerous individual parts that are built and assembled by a combination of processes called manufacturing
- Manufacturing is making products



制造简史 1

Period	Dates	Metals and casting	Various materials and composites	Forming and shaping	Joining	Tools, machining and manufacturing systems
	Before 4000 B.C.	Gold, copper, meteoric iron	Earthenware, glazing, natural fibers	Hammering		Tools of stone, flint, wood, bone, ivory, composite tools
Egypt: ~ 3100 B.C. to ~ 300 B.C. Greece: ~ 1100 B.C. to ~ 146 B.C. Roman Empire: ~ 500 B.C. to 476 A.D.	4000-3000 B.C.	Copper casting, stone and metal molds, lost-wax process, silver, lead, tin, bronze		Stamping, jewelry	Soldering (Cu-Au, Cu-Pb, Pb-Sn)	Corundum (alumina, emery)
	3000-2000 B.C.	Bronze casting and drawing, gold leaf	Glass beads, potter's wheel, glass vessels	Wire by slitting sheet metal	Riveting, brazing	Hoe making, hammered axes, tools for ironmaking and carpentry
	2000-1000 B.C.	Wrought iron, brass				
	1000-1 B.C.	Cast iron, cast steel	Glass pressing and blowing	Stamping of coins	Forge welding of iron and steel, gluing	Improved chisels, saws, files, woodworking lathes
	1-1000 A.D.	Zinc, steel	Venetian glass	Armor, coining, forging, steel swords		Etching of armor
Middle Ages: ~ 476 to 1492. Renaissance: 14th to 16th centuries.	1000-1500	Blast furnace, type metals, casting of bells, pewter	Crystal glass	Wire drawing, gold- and silversmith work		Sandpaper, windmill-driven saw
	1500-1600	Cast-iron cannon, tinplate	Cast plate glass, flint glass	Water power for metalworking, rolling mill for coinage strips		Hand lathe for wood
	1600-1700	Permanent-mold casting, brass from copper and metallic zinc	Porcelain	Rolling (lead, gold, silver), shape rolling (lead)		Boring, turning, screw-cutting lathe, drill press

制造简史 2

Period	Dates	Metals and casting	Various materials and composites	Forming and shaping	Joining	Tools, machining and manufacturing systems
Industrial Revolution: ~ 1750 to 1850	1700-1800	Malleable cast iron, crucible steel (iron bars and rods)		Extrusion (lead pipe), deep drawing, rolling		
	1800-1900	Centrifugal casting, Bessemer process, electrolytic aluminum, nickel steel, babbitt, galvanized steel, powder metallurgy, open-hearth steel	Window glass from slit cylinder, light bulb, vulcanization, rubber processing, polyester, styrene, celluloid, rubber extrusion, molding	Steam hammer, steel rolling, seamless tube, steel-rail rolling, continuous rolling, electroplating		Shaping, milling, copying lathe for gunstocks, turret lathe, universal milling machine, vitrified grinding wheel
WWI	1900-1920		Automatic bottle making, bakelite, borosilicate glass	Tube rolling, hot extrusion	Oxyacetylene; arc, electrical-resistance, and thermit welding	Geared lathe, automatic screw machine, hobbing, high-speed steel tools, aluminum oxide and silicon carbide (synthetic)
WWII	1920-1940	Die casting	Development of plastics, casting, molding, polyvinyl chloride, cellulose acetate, polyethylene, glass fibers	Tungsten wire from metal powder	Coated electrodes	Tungsten carbide, mass production, transfer machines
	1940-1950	Lost-wax process for engineering parts	Acrylics, synthetic rubber, epoxies, photosensitive glass	Extrusion (steel), swaging, powder metals for engineering parts	Submerged arc welding	Phosphate conversion coatings, total quality control
	1950-1960	Ceramic mold, nodular iron, semiconductors, continuous casting	Acrylonitrile-butadiene-styrene, silicones, fluorocarbons, polyurethane, float glass, tempered glass, glass ceramics	Cold extrusion (steel), explosive forming, thermochemical processing	Gas metal arc, gas tungsten arc, and electroslag welding; explosion welding	Electrical and chemical machining, automatic control.

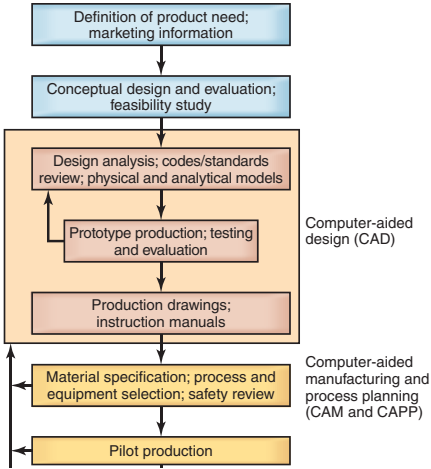
制造简史 3

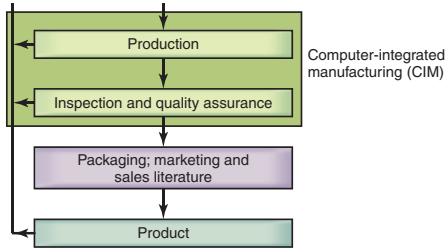
Period	Dates	Metals and casting	Various materials and composites	Forming and shaping	Joining	Tools, machining and manufacturing systems
	1960-1970	Squeeze casting, single-crystal turbine blades	Acetals, polycarbonate, cold forming of plastics, reinforced plastics, filament winding	Hydroforming, hydrostatic extrusion, electroforming	Plasma-arc and electron-beam welding, adhesive bonding	Titanium carbide, synthetic diamond, numerical control, integrated circuit chip
Space Age	1970-1990	Compacted graphite, vacuum casting, organically-bonded sand, automation of molding and pouring, rapid solidification, metal-matrix composites, semi- solid metalworking, amorphous metals, shape-memory alloys	Adhesives, composite materials, semiconductors, optical fibers, structural ceramics, ceramic-matrix composites, biodegradable plastics, electrically-conducting polymers	Precision forging, isothermal forging, superplastic forming, dies made by computer-aided design and manufacturing, net-shape forging and forming, computer simulation	Laser beam, diffusion bonding (also combined with superplastic forming), surface-mount soldering	Cubic boron nitride, coated tools, diamond turning, ultraprecision machining, computer-integrated manufacturing, industrial robots, machining and turning centers, flexible manufacturing systems, sensor technology, automated inspection, computer simulation and optimization
Information Age	1990-2000	Rheocasting, computer-aided design of molds and dies, rapid tooling	Nanophase materials, metal foams, high-temperature superconductors. machinable ceramics, diamond-like carbon	Additive manufacturing, rapid tooling, environmentally-friendly metalworking fluids	Friction stir welding, lead-free solders, laser butt-welded (tailored) sheet-metal blanks	Micro- and nanofabrication, LIGA, dry etching, linear motor drives, artificial neural networks, six sigma
	2000-2010s	TRIP and TWIP steels	Carbon nanotubes, graphene	Single point incremental forming, hot stamping, electrically assisted forming	linear friction welding	Digital manufacturing, three-dimensional computer chips, blue-arc machining, soft lithography, flexible electronics

产品设计与并行工程

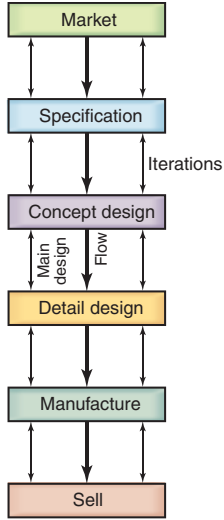
- Product design involves the creative and systematic prescription 规定 of the shape and characteristics of a product to achieve specified objectives, while simultaneously satisfying several important constraints.
- Design is a critical activity, because it has been estimated that as much as 80% of the cost of product development and manufacturing is determined by the decisions made in the initial stages of design.
- Innovative approaches are essential in successful product design, as 同样也 are clearly specified functions and a clear statement of the performance expected of the product.
- The market for a product, which may be new or a modified version of an existing product, and its anticipated use or uses, also must be clearly defined at this stage.
- This aspect also involves the assistance of market analysts and sales personnel 销售人员 who will bring valuable and timely input to the manufacturer, especially regarding market needs and trends.

产品传统设计制造流程


Definition of product need; marketing information
Conceptual design and evaluation; feasibility study
Design analysis; codes/standards review; physical and analytical models
Prototype production; testing and evaluation
Production drawings; instruction manuals
Computer-aided design (CAD)
Material specification; process and equipment selection; safety review
Pilot production
Computer-aided manufacturing and process planning (CAM and CAPP)


Production
Inspection and quality assurance
Packaging; marketing and sales literature
Product
Computer-integrated manufacturing (CIM)

并行工程中的设计制造


Market
Specification
Iterations
Concept design
Main design
Flow
Detail design
Manufacture
Sell

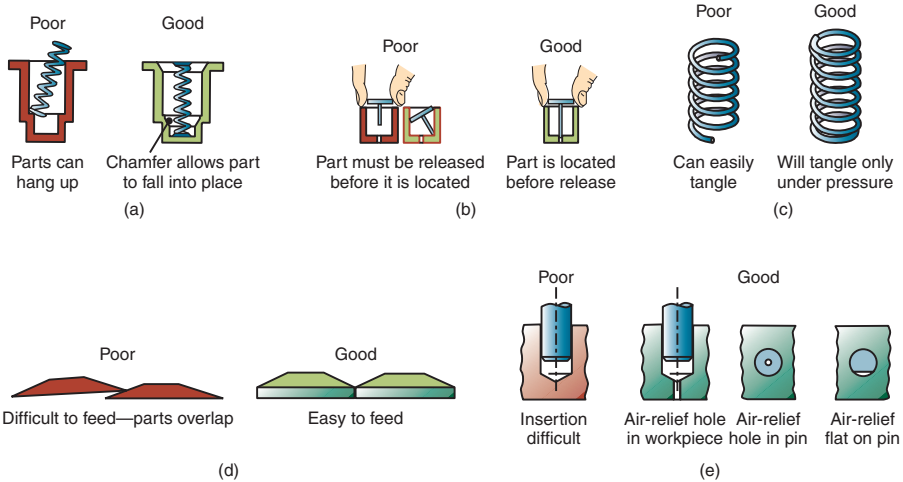
面向制造的设计 Design for Manufacture

- Design for manufacture (DFM) is a comprehensive approach to integrating the design process with production methods, materials, process planning, assembly, testing, and quality assurance.
- DFM requires a fundamental understanding of
  - (a) the characteristics and capabilities of materials, manufacturing processes, machinery, equipment, and tooling, and
  - (b) variability in machine performance, dimensional accuracy and surface finish of the workpiece, processing time, and the effect of processing methods employed on product quality.
- Establishing quantitative relationships is essential in order to be able to analyze and optimize a design, for ease of manufacturing and assembly at the lowest cost.

面向装配的设计 Design for assembly

- The concepts of design for assembly (DFA), design for manufacture and assembly (DFMA), and design for disassembly (DFD) are all important considerations in manufacturing.
- Methodologies and computer software are available for design for assembly, utilizing 3-D conceptual designs and solid models.
- Subassembly, assembly, and disassembly times and costs are minimized, while maintaining product integrity and performance.
- A product which is easy to assemble is usually also easy to disassemble.

易于装配的设计实例


Poor
Good
Parts can hang up
Chamfer allows part to fall into place
(a)
Poor
Good
Part must be released before it is located
Part is located before release
(b)
Poor
Good
Can easily tangle
Will tangle only under pressure
(c)
Poor
Good
Difficult to feed—parts overlap
Easy to feed
(d)
Poor
Good
Insertion difficult
Air-relief hole in workpiece
Air-relief hole in pin
Air-relief flat on pin
(e)

面向服务的设计 Design for Service

- Products often have to be disassembled, to varying degrees, in order to service them and, if necessary, repair them.
- The design should take into account the concept that, for ease of access, components that are most likely to be in need of servicing be placed, as much as possible, at the outer layers of the product.
- This methodology can be appreciated by anyone who has had the experience of servicing machinery.



Waste from manufacturing operations

1. Chips from machining and trimmed materials from sheet forming, casting, and molding operations
2. Slag from foundries and welding operations
3. Additives in sand used in sand-casting operations
4. Hazardous waste and toxic materials used in various products
5. Lubricants and coolants in metalworking and machining operations
6. Liquids from such processes as heat treating and plating
7. Solvents from cleaning operations
8. Smoke and pollutants from furnaces and gases from burning fossil fuels.

工业废弃物及回收

- 塑料污染问题逐渐成为仅次于气候变化的全球第二大焦点环境问题，我国塑料废弃物年产生量达 6000 万吨，2019 年国内废塑料回收量约 1890 万吨
- 2019 年，我国废纸回收量约 5244 万吨，废纸回收率为 49.0%，废纸利用率为 58.3%

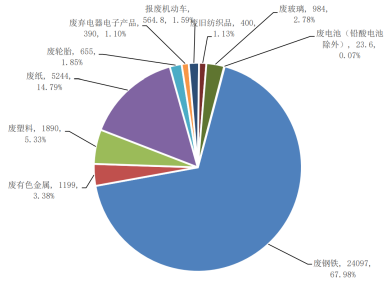


图 1: 2019 年我国主要品种再生资源回收量及占比情况

环保意识设计 Environmentally Conscious Design（ECD）

- The term sustainable design and manufacturing 可持续设计与制造 has become in common usage in all industrial activities, with major emphasis on design for the environment (DFE).
- Also called environmentally conscious design and manufacturing 具有环境意识的设计与制造 and green design, this approach considers all possible adverse environmental impacts of materials, processes, operations, and products, so that they can all be taken into account at the earliest stages of their design and production.

面向资源回收的设计 design for recycling (DFR)

- Recycling may involve one of two basic activities:
  - Biological cycle 生物循环: Organic materials degrade naturally, and in the simplest version of a biological cycle, they lead to new soil that can sustain life. Thus, product design involves the use of organic materials, as well as ensuring that products function well for their intended life and can then be safely disposed of. 纸袋、布袋替代塑料袋
  - Industrial cycle 工业循环: The materials in the product are recycled and reused continuously. To demonstrate the economic benefits of this approach, it has been estimated that producing aluminum from scrap, instead of from bauxite ore 铝土矿, reduces production costs by as much as 66% and reduces energy consumption and pollution by more than 90%. 废铜、废铁、废铝回收

Guidelines for Sustainable Design and Manufacturing

1. Reduce waste of materials, by refining product design, reducing the amount of materials in products, and select manufacturing processes that minimize scrap (such as forming instead of machining).
2. Reduce the use of hazardous materials in products and processes.
3. Investigate manufacturing technologies that make environmentally friendly and safe products and by-products.
4. Make improvements in methods of recycling, waste treatment, and reuse of materials.
5. Minimize energy use; whenever possible, encourage the use of renewable sources of energy. Select materials can have a major impact on the latent energy in products.
6. Encourage recycling by using materials that are a part of either industrial or biological cycling, but not both in the same product. Ensure proper handling and disposal of all waste of materials that are used in products, but are not appropriate for industrial or biological cycling.

选材 Selection of Materials

- Ferrous metals: Carbon, alloy, stainless, and tool and die steels
- Nonferrous metals: Aluminum, magnesium, copper, nickel, titanium, superalloys, refractory metals, beryllium 铍, zirconium 锆, low-melting-point alloys, and precious metals 贵金属
- Plastics (polymers): Thermoplastics, thermosets, and elastomers
- Ceramics, glasses, glass ceramics 玻璃陶瓷，如昆仑玻璃, graphite, diamond, and diamond-like materials
- Composite materials: Reinforced plastics and metal-matrix and ceramic-matrix composites
- Nanomaterials
- Shape-memory alloys (smart materials), amorphous alloys 非晶合金, semiconductors, and superconductors 超导体

选材需要考虑的因素

- Properties of Materials. Mechanical properties of interest in manufacturing generally include strength, ductility, hardness, toughness, elasticity, fatigue, and creep resistance. Physical properties are density, specific heat, thermal expansion and conductivity, melting point, and electrical and magnetic properties. Chemical properties include oxidation, corrosion, degradation, toxicity, and flammability.
- Material Substitution in Products
- Availability The economic aspect of material selection is as important as technological considerations. Geopolitics (地缘政治) also must thus be a consideration, particularly during periods of global instability or hostility.
- Service Life

选材的多样性


Steel pistons with friction-welded skirt and crown and chrome/ molybdenum disulfide coated piston ring
Lexan windshield
Silicon microprocessors with gold-plated connectors
Fiberglass hood
Copper tube, Aluminum fin coolers
High strength steel bolts

选工艺 Selection of Manufacturing Processes

- There is often more than one method that can be employed to produce a part from a given material
- The selection of a particular manufacturing process 工艺 or, more often, sequence of processes 工序, depends on the geometric features of the parts to be made, including the dimensional tolerances 尺寸公差 and surface texture 表面纹理，如可用粗糙度表征 required, and on numerous factors pertaining to the particular workpiece material and its manufacturing properties

Difficulties for different process

- Flat parts and thin cross sections can be difficult to cast
- Complex parts generally cannot be shaped easily and economically by such metalworking techniques 金属工艺 as forging, whereas, depending on part size and level of complexity, the parts may be precision cast, fabricated and assembled from individual pieces, or produced by powder-metallurgy techniques
- Dimensional tolerances and surface finish in hot-working operations are not as fine as those obtained in operations performed at room temperature (cold working), because of the dimensional changes, distortion, warping 扭曲, and surface oxidation due to elevated temperatures

常用的制造工艺

1. Casting 铸造: Expendable mold and permanent mold
2. Forming and shaping 成形: Rolling, forging, extrusion, drawing, sheet forming, powder metallurgy, and molding
3. Machining 切削: Turning, boring, drilling, milling, planing 龙门刨, shaping 牛头刨, broaching; grinding; ultrasonic machining; chemical, electrical, and electrochemical machining; and high-energy-beam machining
4. Joining 连接: Welding 焊接, brazing 钎焊, soldering 锡焊, diffusion bonding, adhesive bonding, and mechanical joining
5. Finishing 精整加工: Honing, lapping, polishing, burnishing, deburring 去毛刺, surface treating, coating, and plating
6. Microfabrication and nanofabrication 微纳制造: producing parts with dimensions at the micro and nano levels; fabrication of MEMS and NEMS, typically involving processes such as lithography 光刻, micromachining, etching, LIGA, and various specialized processes

计算机集成制造 Computer-integrated Manufacturing

- *Computer-integrated manufacturing (CIM)*, as the name suggests, integrates the software and the hardware needed for computer graphics, computer-aided modeling, and computer-aided design and manufacturing activities, from initial product concept through its production and distribution in the marketplace

Elements in CIM 1/5

1. Computer numerical control (CNC). First implemented in the early 1950s, this is a method of controlling the movements of machine components by direct insertion of coded instructions in the form of numerical data.
2. Adaptive control (AC). The processing parameters in an operation are automatically adjusted to optimize the production rate and product quality and to minimize manufacturing costs.
3. Industrial robots. Introduced in the early 1960s, industrial robots have rapidly been replacing humans, especially in operations that are repetitive, dangerous, and boring. As a result, variability in product quality decreases and productivity is improved. Robots are particularly effective in assembly operations; *intelligent robots* have been developed with sensory perception capabilities and movements that simulate those of humans. Recent innovations involve *cobots*, which are designed to work and interact with humans.

Elements in CIM 2/5

1. Automated materials handling. Computers have made possible highly efficient handling of materials and parts in various stages of completion (work in progress), as in moving a part from one machine to another, and then to points of inspection, to inventory, and, finally, to shipment.
2. Automated assembly systems. These systems have been developed to replace assembly by human operators, although humans still have to perform some of the operations. Depending on the type of product, assembly costs can be high; thus, products must be designed such that they can be assembled more easily and faster by automated machinery.
3. Computer-aided process planning (CAPP). By optimizing process planning, this system is capable of improving productivity, product quality, and consistency, thus reducing costs. Functions such as cost estimating and monitoring work standards (time required to perform a certain operation) are also incorporated into the system.

Elements in CIM 3/5

1. Group technology (GT). The concept behind group technology is that numerous parts can be grouped and produced by classifying them into families according to similarities in (a) design and (b) the manufacturing processes employed to produce them. In this way, part designs and processing plans can be standardized, and new parts, based on similar parts made previously, can be produced efficiently and economically.
2. Just-in-time production (JIT) 准时化生产. The principle behind JIT is that (a) supplies of raw materials and parts are delivered to the manufacturer just in time to be used, (b) parts and components are produced just in time to be made into subassemblies, and (c) products are assembled and finished just in time to be delivered to the customer. As a result, inventory carrying costs are minimal, defects in components are detected right away, productivity is increased, and high-quality products are made and at low cost.

Elements in CIM 4/5

1. Cellular manufacturing (CM) 单元式制造. This system utilizes workstations that consist of a number of manufacturing cells, each containing various production machines, all controlled by a central robot, with each machine performing a specific operation on the part, including inspection.
2. Flexible manufacturing systems (FMS). These systems integrate manufacturing cells into a large production facility, in which all cells are interfaced with a central computer. Although very costly, flexible manufacturing systems are capable of producing parts efficiently (although in relatively small quantities, because hard automation is still most efficient for mass production) and of quickly changing manufacturing sequences required for making different types of parts. Flexibility enables these systems to meet rapid changes in market demand for all types of products.

Elements in CIM 5/5

1. Expert systems (ES). Consisting basically of complex computer programs, these systems have the capability of performing a variety of tasks and solving difficult real-life problems (much as human experts would), including expediting the traditional iterative process.
2. Artificial intelligence (AI). Computer-controlled systems are capable of learning from experience and of making decisions that optimize operations and minimize costs, ultimately replacing human intelligence.

质量保证与质量管理

- Product quality is one of the most critical considerations in manufacturing, because it directly influences customer satisfaction, thus playing a crucial role in determining a product's success in the marketplace
- *quality must be built into the product* The objective is to *control processes, not products*
- Producing and marketing defective products can be very costly to the manufacturer, with costs varying by orders of magnitude

Stage	Relative cost of repair
When part is being made	1
Subassembly of the product	10
Assembly of the product	100
Product at the dealership	1000
Product at the customer	10,000

Quality Standards

- *ISO 9000 and QS 9000* have permanently influenced the manner in which companies conduct business in world trade
- A company's registration for these standards, which is a *quality process certification* and *not a product certification*, means that the company conforms to consistent practices as specified by its own quality system

Human-factors Engineering 人因工程

- It emphasizes ergonomics 人体工程学, defined as the study of how a workplace and the machinery and equipment in it, can best be designed and arranged for comfort, safety, efficiency, and productivity
  - (a) a mechanism that is difficult to operate manually, causing injury to the employee
  - (b) a poorly designed keyboard that causes pain to the user's hands and arms during its normal use (repetitive stress syndrome)
  - (c) a control panel on a machine which is difficult to reach or use safely and comfortably

Product Liability 产品责任/产品风险

- Designing and manufacturing safe products is an essential responsibility of the manufacturer
- All those involved with product design, manufacture, and marketing must fully recognize the consequences of a product's failure, including failure due to foreseeable misuse of the product
- A product's malfunction or failure can cause bodily injury or even death, as well as financial loss to an individual, a bystander, or an organization

精益生产 Lean Production

- *Lean production* is a methodology that involves thorough assessment of each activity of a company.
- Its basic purpose is to minimize waste at all levels, and calling for the elimination of unnecessary operations that do not provide any added value to the product being made.
- This approach, also called *lean manufacturing*, identifies all of a manufacturer's activities and optimizes the processes used in order to maximize added value.
- Lean production does not necessarily require cutting back on a company's physical or human resources. It aims at continually improving efficiency and profitability by removing all waste in the company's operations and dealing with any problems as soon as they arise.

制造趋势 1/2

- 产品多样性、复杂性继续加强
- 产品生命周期不断缩短
- 市场全球化、竞争日益增加
- 顾客要求越来越高
- 新材料、替代材料不断发展，促进循环利用
- 切削精度进一步增加
- 优先采用经济、环境友好的制造方法，能量管理变得越来越重要
- 越来越注重减重，交通运输领域为提高燃油经济性，钛、镁、铝以及树脂复材应用越来越广
- 产品设计阶段，研究材料工艺参数对产品完整性影响的预测模型越来越准确
- 超精密制造、微纳制造不断发展，制造已趋于原子尺度

制造趋势 2/2

- 计算机模拟、建模、控制策略正应用于预测制造、设计、性能及维护的各个领域
- 增材制造应用范围不断扩大
- 制造工艺及生产系统的优化效果更加可靠
- 精益生产与信息技术被广泛使用
- 制造活动被视为各部分相互关联的大系统
- 将质量控制融入生产的每一个环节已成为共识
- 优化组织资源，消除或减少浪费，提高生产率
- 软件应用扩展到了生产制造的每个环节
- 各类型传感器集成到机器